THE UNIVERSITY OF BRITISH COLUMBIA Building Operations Facilities	**Policy No.:** **11**	**Last Revision:** **10/10/2024**
	Responsible Executive: Lee McCormick	
Title: **Emergency Generator Testing Procedure**		
Background & Purposes: There are many Emergency Generators across campus which provide power to life safety and critical equipment which are housed within these buildings. As per the CSA-282 code these generators need to be tested monthly for proper operation and transfer.		

1. General

Emergency Generators have the requirement to be functionally tested once per month. The monthly testing must comply with the Canadian code (CSA282-15) for the emergency electrical supply for buildings testing procedures outlined below:

- Simulate a failure of the normal electrical supply to the building
- Verify the battery charger current output increases during cranking
- Operate the system under at least 40% of rated load for at least 60 minutes
- Operate all automatic transfer switches under load
- Inspect brush operation for sparking (if applicable)
- Inspect for motor bearing seal leakage
- Inspect for proper operation of auxiliary equipment, EG radiator shutter control, coolant pumps, fuel transfer pumps, oil coolers, and engine room ventilation systems
- Record all instrument readings in the log onsite and verify that they are normal
- Drain the exhaust system condensate trap
- Correct all defects found during the tests

The generators are tested according to the generator testing schedule created and maintained through the Building Operations planning department. The Operations department is responsible for executing the schedule.

2. Detailed Monthly Procedure

1. Receive and follow the schedule outlined by the planning coordinator.
2. Travel to site for the generator test.
3. Contact TELUS (604) 731-4126 and let them know to put the system onto test for the allotted time.
4. Perform a pre-start system inspection on the generator, and associated equipment. Refer to the Emergency Generator Information sheets for the building specific information to follow.
5. Shut down any power to the equipment onsite that is affected by the transfer of power.
6. Complete the pre-check list for generator and fuel tank on the Planon work order.
7. Commence with starting the generator from the transfer switch location
8. Operate all the building transfer switches associated with the generator and run it for at least 60 minutes.
9. Check back at the generator to confirm that the unit is running properly and record the coolant temperature and oil pressure.
10. Check the building fire alarm panel, acknowledge any troubles or supervisory codes. Reset the fire panel if needed. Perform this action only once, and if the panel does not reset contact FLS for further investigation.
11. Record the amperage readings on the gauges at the generator 30 minutes into the test.
12. After one hour, transfer the load back to the building and turn the generator off.
13. Call TELUS back and let them know to release the test on the system.
14. Record all findings in logbook onsite and let your supervisor know of any issues found.